

TARIK AKARLI

[portfolio](#)

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🐙 [github](#)

Education

Durham University

Sep 2024 – May 2028

Master of Science in Mathematics and Computer Science

Durham, UK

Predicted 1:1. Relevant modules of study and grades can be found [here](#).

Newcastle Sixth Form College

Sep 2022 – Jun 2024

A levels: Further Mathematics (A), Mathematics (A*), Computer Science (A), Physics(A)*

Newcastle upon Tyne, UK

Experience and Leadership

NEDES

Sep 2024 – Present

Maths Tutor

Newcastle upon Tyne, UK

- Tutor primary and secondary school students in a classroom setting, clarifying mathematical topics from the core curriculum.
- Manage a classroom environment of up to 6 students, fostering student engagement and analytical problem-solving skills.

Dialogue Society

Dec 2022 – Present

Volunteer and Youth Platform Leader

North East, England

- Collaborate with a team to organise and facilitate events aimed at fostering inter-community dialogue.
- Acted as Master of Ceremonies (MC) for 4 events, engaging diverse audiences of up to 200 and facilitating programme schedules, demonstrating strong public speaking and audience management skills.

Mentor Wise UK

Sep 2023 – Present

Mentor

North East, England

- Fostered a supportive and confidential environment by organising weekly meetings, enabling 7 mentees to build self-confidence and effectively address personal or academic challenges.
- Assisted mentees in developing crucial life skills, including goal-setting, time management, and effective study techniques.

Projects

Portfolio Optimizer | Python, Pandas, NumPy

Jan 2025

- Developed a quantitative tool in Python to construct optimal investment portfolios using Markowitz's Modern Portfolio Theory.
- Implemented mean-variance optimisation to compute the efficient frontier for a custom basket of assets, identifying the optimal risk-return trade-offs.
- Utilised NumPy and Pandas for data manipulation and mathematical calculations, and Matplotlib for visualising the efficient frontier and capital allocation line.

High-Frequency Trading Simulator | Python, NumPy, Pandas

Apr 2025

- Built a simulation environment in Python to model and back-test high-frequency trading strategies on historical limit order book data.
- Engineered market-making strategy, evaluating its profitability and performance metrics like Sharpe ratio and max draw-down.

Monte Carlo Pricing of an Exotic Option | Python, Pandas, NumPy

Jul 2025

- Built a Streamlit app to Monte Carlo-price discrete arithmetic Asian options under GBM, with dividends and call/put support.
- Added variance reduction (antithetic, moment matching, control variate). Produced clear diagnostics and exports: convergence/distribution plots, sample paths, CSV

Any projects and their applications and any academic info can be found on my portfolio [here](#).

Technical Skills

Languages: Python, R, HTML/CSS, JavaScript, SQL

Libraries and Frameworks: Pandas, NumPy, Matplotlib

Technologies and Tools: Linux, GitHub, LaTeX, Excel

Spoken Languages: English(Fluent), Turkish(Fluent)